

# Polar Curves and Derivatives



## Converting between polar and rectangular coordinates

- 1 Convert the polar point  $(4, \frac{\pi}{3})$  to rectangular coordinates.
  - 2 Convert the polar equation  $r = 4\cos\theta$  to rectangular form, and identify the curve.
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## Identifying and sketching common polar curves

- 3 Identify the type of curve for  $r = 2 + 2\sin\theta$ , and state its key feature.
  - 4 Identify the type of curve for  $r = \cos(3\theta)$ .
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## The slope of a polar curve

- 5 State the formula for  $\frac{dy}{dx}$  of a polar curve  $r = f(\theta)$ , using  $x = r\cos\theta$  and  $y = r\sin\theta$ .
  - 6 Find  $\frac{dy}{dx}$  for  $r = 2\theta$  at  $\theta = \frac{\pi}{2}$ .
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## Finding horizontal and vertical tangents

- 7 For the circle  $r = 4\cos\theta$ , find the value(s) of  $\theta$  in  $[0, \pi]$  where the tangent line is horizontal (where  $y = r\sin\theta$  has a critical point).
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## Symmetry of polar curves

- 8 Explain how to test whether a polar curve  $r = f(\theta)$  is symmetric about the polar axis (the x-axis).
  - 9 Test whether  $r = 3\sin\theta$  is symmetric about the y-axis (replace  $\theta$  with  $\pi - \theta$ ).
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## More polar-to-rectangular conversions

- 10 Convert the rectangular point  $(3, 3\sqrt{3})$  to polar coordinates ( $r > 0, 0 \leq \theta < 2\pi$ ).
- 11 Convert the rectangular equation  $y = x$  to polar form.

- 12 Convert the polar equation  $r \sin \theta = 4$  to rectangular form, and identify the curve.
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### More slope-of-a-polar-curve practice

- 13 Find  $\frac{dy}{dx}$  for the cardioid  $r = 1 + \sin \theta$  at  $\theta = 0$ .
- 14 Find  $\frac{dy}{dx}$  for the circle  $r = 5$  (constant) at any angle  $\theta$ , and interpret the result using the fact that a circle centered at the origin has radial symmetry.
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### Tangent lines at the pole

- 15 Find the values of  $\theta$  where the cardioid  $r = 1 - \cos \theta$  passes through the pole ( $r = 0$ ), and explain why these give the tangent line directions at the pole.
- 16 Find the values of  $\theta \in [0, 2\pi)$  where the rose  $r = \cos(2\theta)$  passes through the pole.
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### Classifying polar curves by equation type

- 17 Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):
- |                          |                          |
|--------------------------|--------------------------|
| a $r = 5$                | b $r = 3 + 3\cos \theta$ |
| c $r = 1 + 3\sin \theta$ | d $r = 4\sin(5\theta)$   |
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### Number of petals in rose curves

- 18 State the rule for the number of petals of  $r = a\cos(n\theta)$  or  $r = a\sin(n\theta)$  depending on whether  $n$  is odd or even.
- 19 How many petals does  $r = 2\sin(4\theta)$  have? How many does  $r = 5\cos(7\theta)$  have?
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### Graphing polar curves point-by-point

- 20 Complete a short table of  $(\theta, r)$  values for  $r = 2 + 2\cos \theta$  at  $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ , and use it to describe the cardioid's shape.
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## Concavity considerations for polar curves

- 21 Explain, in terms of  $\frac{dy}{dx}$ , what a 'vertical tangent' means for a polar curve, and why it corresponds to  $\frac{dx}{d\theta} = 0$  (with  $\frac{dy}{d\theta} \neq 0$ ).